Geo_Tracker_app: Android GPS Tracking, Geofencing and Navigation System

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1. Introduction

Geo_Tracker_app is an Android mobile application developed using Java that enables users to track their current location, save important geographical points, receive geofence-based notifications, and navigate to saved locations using Google Maps. The application demonstrates the use of Android location-based services, local storage using Room database, and user-friendly interface design.

2. Objectives

- Track the user's real-time GPS location.
- Allow users to add and save custom locations.
- Store location data using Room database.
- Implement geofencing around saved locations.
- Notify users when entering a geofence.
- Provide navigation using Google Maps.
- Design a simple and intuitive user interface.

3. System Architecture

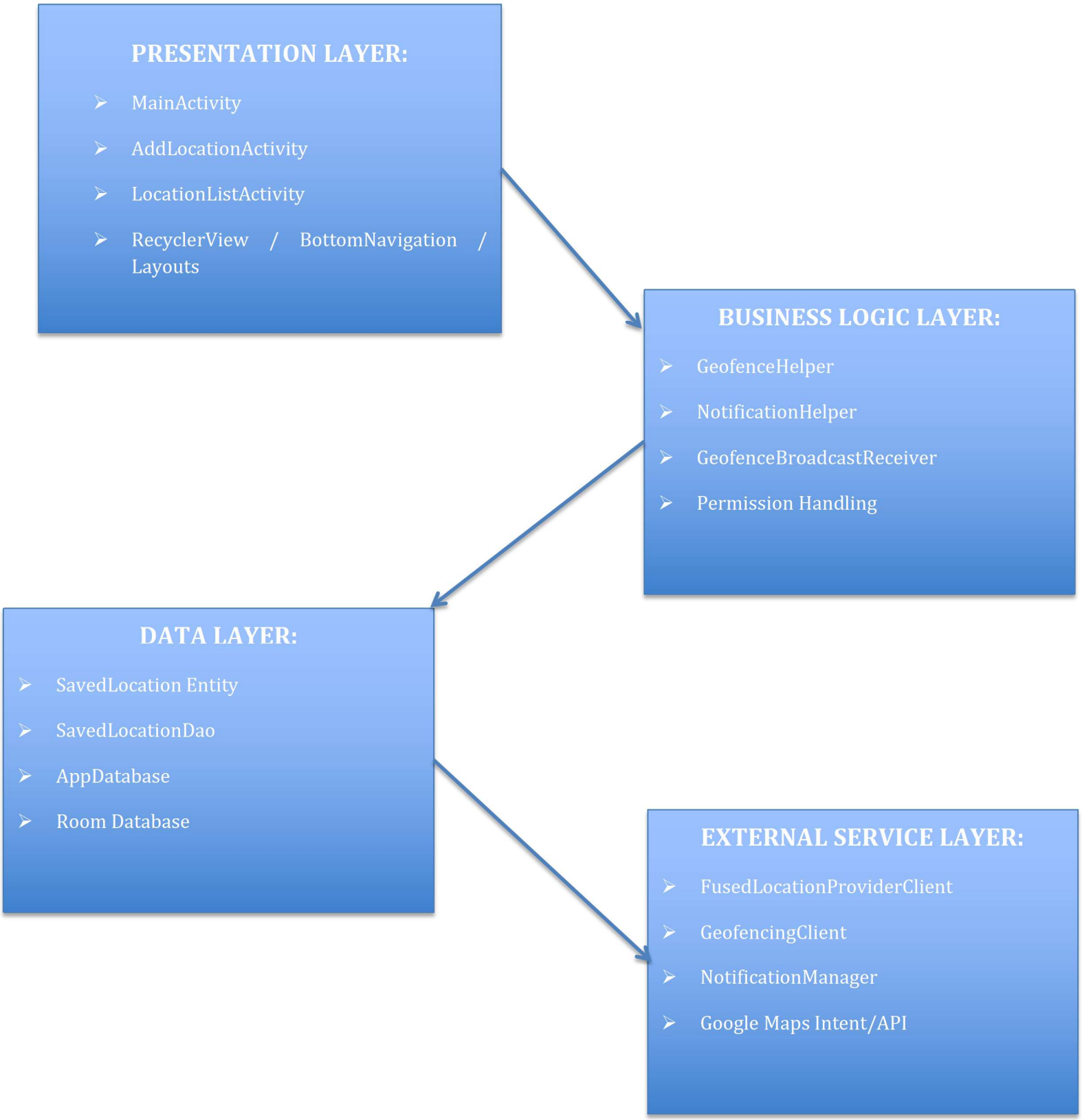
The application follows a layered architecture consisting of:

1. **Presentation Layer:** Activities and UI components.

2. Data Layer: Room Database (SavedLocation, DAO, AppDatabase).

3. Business Logic Layer: GeofenceHelper, NotificationHelper, BroadcastReceiver.

4. External Services: GPS, GeofencingClient, Google Maps.



4. Implementation Details

(a) GPS Tracking: Implemented using FusedLocationProviderClient.

(b) Local Storage: Implemented using Room Database for persistence.

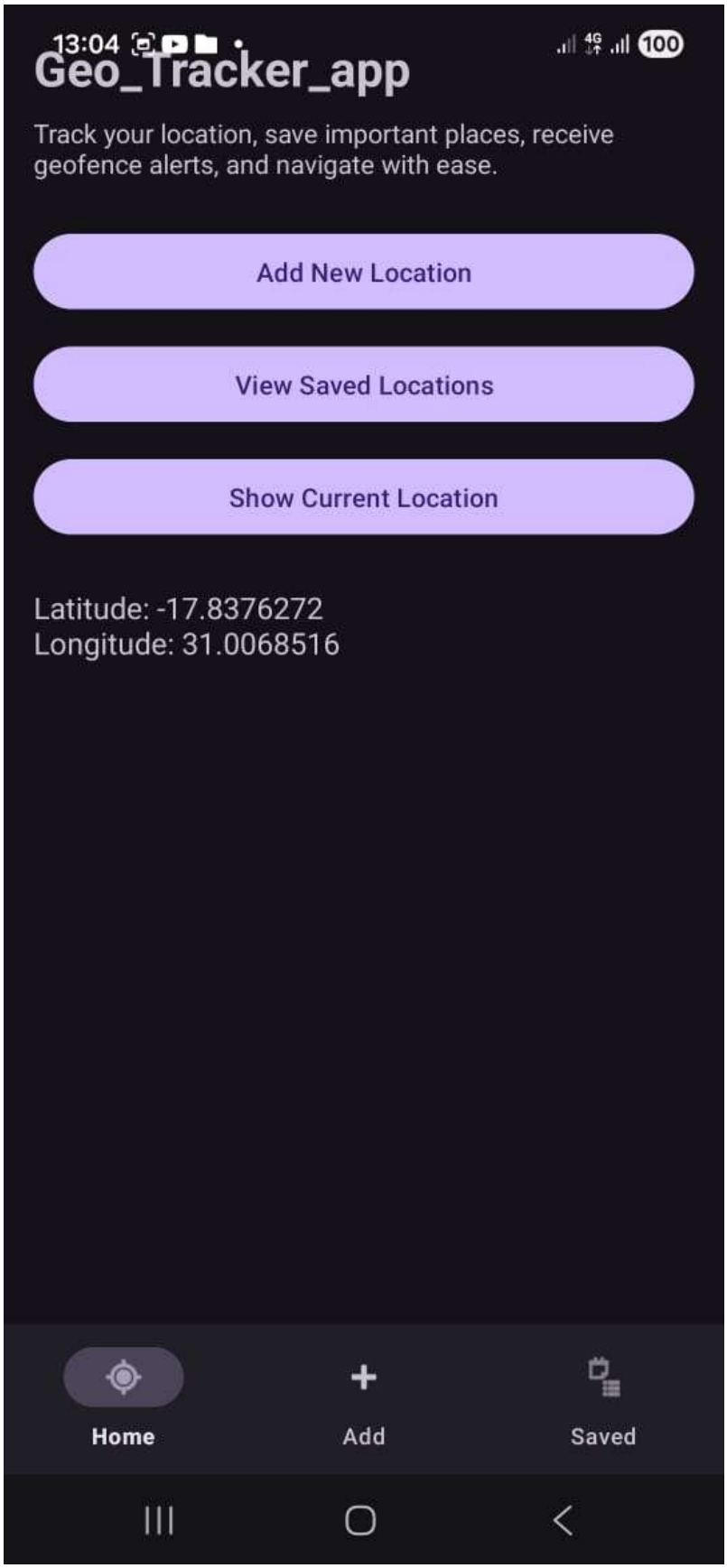
(c) Geofencing: Implemented using GeofencingClient with entry triggers.

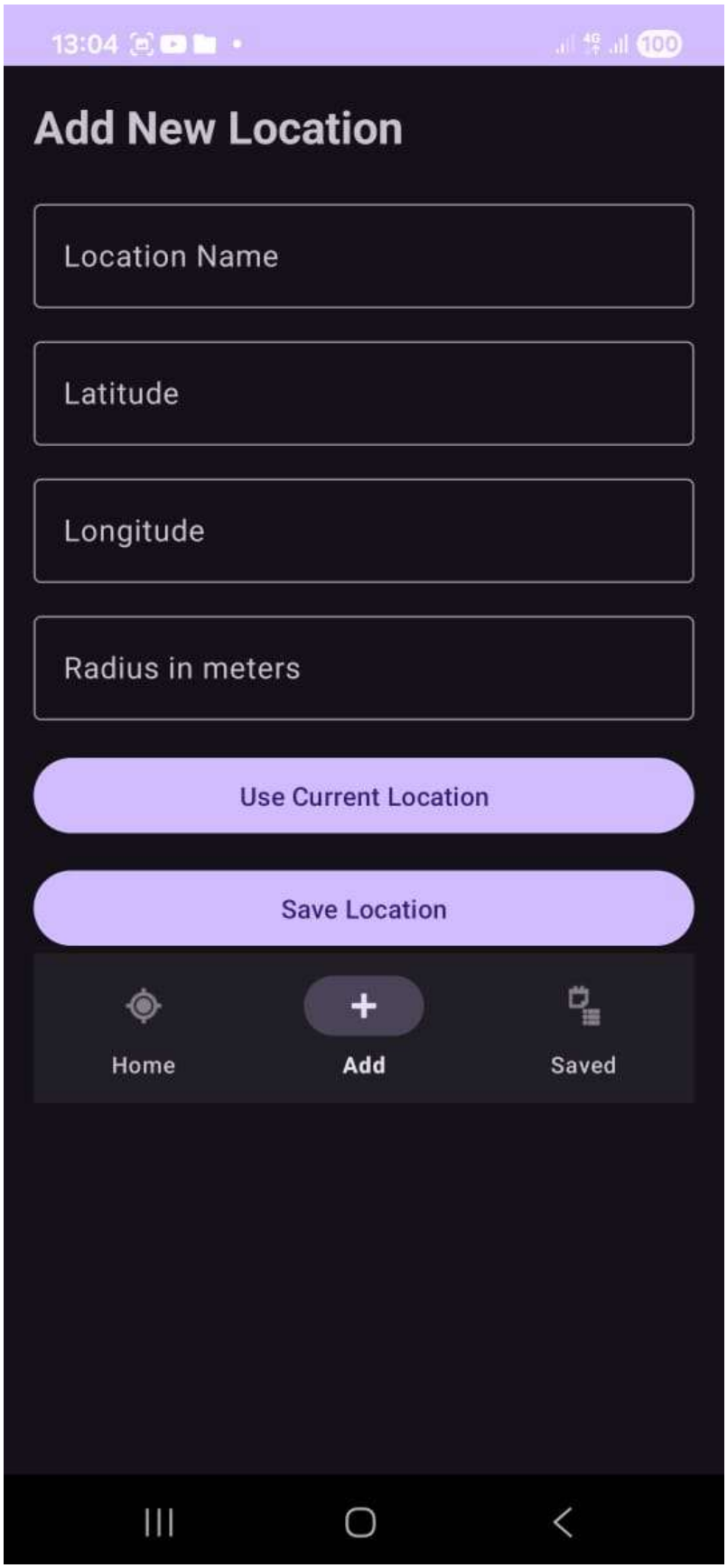
(d) Notifications: Triggered when user enters a geofence.

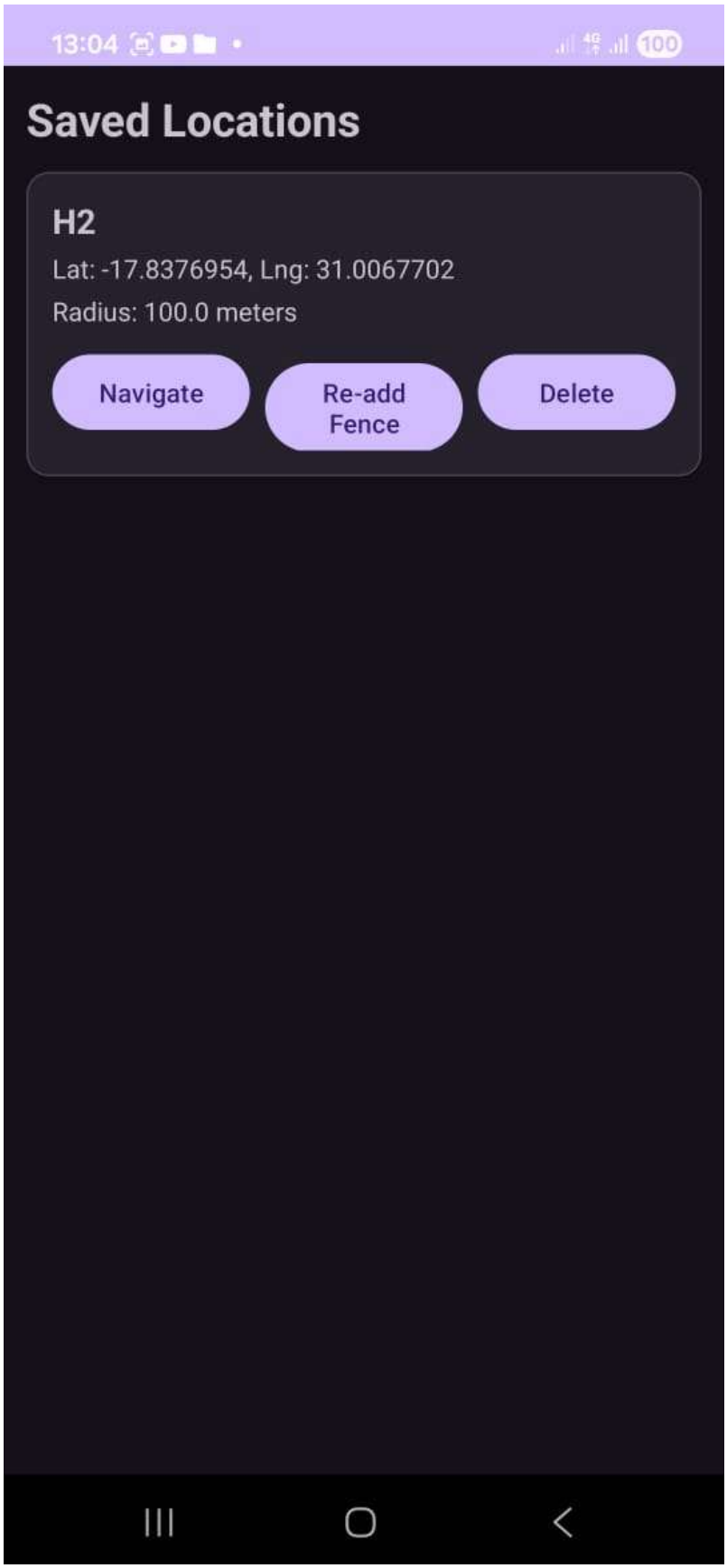
(e) Navigation: Implemented using Google Maps Intent.

5. User Interface Design

The app uses Material Design principles with a bottom navigation bar for easy navigation between screens. RecyclerView is used to display saved locations.


13:04
100
Geo_Tracker_app
Track your location, save important places, receive geofence alerts, and navigate with ease.
Add New Location
View Saved Locations
Show Current Location
Latitude: -17.8376272
Longitude: 31.0068516
Home
Add
Saved


13:04
100
Add New Location
Location Name
Latitude
Longitude
Radius in meters
Use Current Location
Save Location
Home
Add
Saved


13:04
100
Saved Locations
H2
Lat: -17.8376954, Lng: 31.0067702
Radius: 100.0 meters
Navigate
Re-add Fence
Delete

6. Challenges and Solutions

- **Location Unavailable Issue:** Fixed using getCurrentLocation().
- **Background Permission:** Implemented runtime permission handling.
- **Geofence Testing**: Used emulator/mock locations.
- **UI Issues:** Fixed layout and navigation problems.

7. Conclusion

Geo_Tracker_app successfully implements GPS tracking, geofencing, notifications, and navigation features. It demonstrates strong understanding of Android development concepts and meets the assignment requirements.

Source Code is in the respository: